



Class: - B.E.D.S.
Subject: Natural Language Processing Lab

Semester: -VII
A.Y: - 2026-27

Experiment No.2

Aim: To perform stop word removal, stemming and lemmatization for a given text.

Software used: Google, Colab – Python

Theory:

Text Preprocessing

Text preprocessing is the process of converting raw text into a clean and structured format that can be easily processed by a computer. It is one of the most important steps in Natural Language Processing (NLP). The quality of preprocessing directly affects the performance of NLP models. Common preprocessing techniques include tokenization, stop word removal, stemming, and lemmatization. These techniques help remove unnecessary information and improve the efficiency and accuracy of text analysis.

Stop Words

Stop words are the most frequently occurring words in a language, such as the, is, are, a, an, in, on, for, to, etc. These words usually do not contribute much to the meaning of a sentence and therefore are often removed during text preprocessing.

For example,

Sentence:

Ex: There is a pen on the table.

After removing stop words:

Output: pen table

Removing stop words reduces the size of the dataset, decreases processing time, and allows the NLP model to focus on meaningful words. However, stop word removal should not always be performed. In applications such as machine translation, text summarization, and question answering, stop words help preserve the meaning and grammatical structure of sentences.

Advantages of Stop Word Removal

- Reduces dataset size.
- Improves training speed.
- Removes unnecessary words from text.



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- Improves text classification performance.
- Makes search and information retrieval faster.

Stemming

Stemming is the process of reducing a word to its root or stem by removing prefixes or suffixes. It is a simple and fast technique that does not consider the meaning or context of the word. As a result, the generated stem may not always be a valid dictionary word.

Examples

Original Word	Stemmed Word
Playing	Play
Running	Run

Popular Stemming Algorithms

- Porter Stemmer
- Snowball Stemmer
- Lancaster Stemmer

Lemmatization

Lemmatization is the process of converting a word into its **dictionary (lemma) form** by considering its grammatical meaning and context. Unlike stemming, lemmatization produces valid words.

Examples

Original Word	Lemmatized Word
Studies	Study
Running	Run



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Common lemmatization tools include:

- WordNet
- NLTK WordNet Lemmatizer
- spaCy

Advantages

- Produces meaningful dictionary words.
- Improves text understanding.
- Gives better accuracy than stemming.
- Useful in sentiment analysis and machine translation.

Program:

▶ `import nltk`

```
nltk.download('punkt_tab')  
nltk.download('punkt')  
nltk.download('stopwords')  
nltk.download('wordnet')  
nltk.download('omw-1.4')
```

```
▶ import nltk  
from nltk.corpus import stopwords  
from nltk.tokenize import word_tokenize  
from nltk.stem import PorterStemmer, WordNetLemmatizer  
  
# Sample English text  
english_text = "Hello there! How are you doing today? This is a sample text for tokenization, stop word removal, stemming, and lemmatization"  
  
# Tokenize the text  
words = word_tokenize(english_text)  
  
# Initialize the Porter Stemmer and WordNet Lemmatizer  
ps = PorterStemmer()  
lemmatizer = WordNetLemmatizer()  
  
# Remove stop words, perform stemming, and lemmatization  
filtered_words = []  
stemmed_words = []  
lemmatized_words = []  
  
stop_words = set(stopwords.words('english'))
```



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```
for word in words:
    if word.lower() not in stop_words:
        filtered_words.append(word)
        stemmed_word = ps.stem(word)
        stemmed_words.append(stemmed_word)
        lemmatized_word = lemmatizer.lemmatize(word)
        lemmatized_words.append(lemmatized_word)

# Display the results
print("Original Text:")
print(english_text)

print("\nTokenized Words:")
print(words)

print("\nFiltered Words (Stop Words Removed):")
print(filtered_words)

print("\nStemmed Words:")
print(stemmed_words)

print("\nLemmatized Words:")
print(lemmatized_words)
```

Output:

```
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt_tab.zip.
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Package stopwords is already up-to-date!
[nltk_data] Downloading package wordnet to /root/nltk_data...
[nltk_data]   Package wordnet is already up-to-date!
[nltk_data] Downloading package omw-1.4 to /root/nltk_data...
[nltk_data]   Package omw-1.4 is already up-to-date!
True
```

```
... Original Text:
Hello there! How are you doing today? This is a sample text for tokenization, stop word removal, stemming, and lemmatization.

Tokenized Words:
['Hello', 'there', '!', 'How', 'are', 'you', 'doing', 'today', '?', 'This', 'is', 'a', 'sample', 'text', 'for', 'tokenization', ',', 'stop', 'word', 'removal',

Filtered Words (Stop Words Removed):
['Hello', '!', 'today', '?', 'sample', 'text', 'tokenization', ',', 'stop', 'word', 'removal', ',', 'stemming', ',', 'lemmatization', '.']

Stemmed Words:
['hello', '!', 'today', '?', 'sampl', 'text', 'token', ',', 'stop', 'word', 'remov', ',', 'stem', ',', 'lemmat', '.']

Lemmatized Words:
['Hello', '!', 'today', '?', 'sample', 'text', 'tokenization', ',', 'stop', 'word', 'removal', ',', 'stemming', ',', 'lemmatization', '.']
```

Conclusion:

Stopwords play an important role in problems like sentiment analysis, question answering systems, etc. That's why removing stopwords can potentially affect our model's accuracy drastically.